

Jason White | Roboticist

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Employment

Foundation Future Industries

2026 - present

Senior Controls and Software Engineer (C++ & Python)

- o Lead controls and software development for a 22 DOF, 25 actuator tendon driven hand, scaling it from a single motor prototype to a modular control stack
<https://www.youtube.com/watch?v=Nhr7ZnFZYkA&t=1s>
- o End to end deployment of hand hardware bring up to mounting and control on Phantom robot using RL policy whole body control with motion retargeting and inputs from mocap suit
- o Led design decisions on tactile & position sensing, PCB components & architecture, and communication protocol & rate
- o Built full stack codebase and control of simulation and hardware in **eight weeks**
- o Developed calibration routine, system identification routine, and safety control in **three weeks**
- o Created highly accurate open loop position and torque dependent state estimator with a 10 microsecond solve time in **two weeks**

Ghost Robotics

2024 - 2026

Controls and Software Engineer (C++ & Python)

- o Built a whole-body MPC locomotion controller on Pinocchio rigid body dynamics, solving at 1kHz on onboard compute
- o Led sys ID and profile design of next-generation leg actuators and drive train, delivering **6x torque density** (3x peak torque and half the mass)
- o Cut falls & flips on rugged terrain from **35% to 5%** over 20 trials by expanding the state machine and tuning heuristics
- o Developed point-cloud clustering and 2D height-map filtering for terrain-aware locomotion, enabling traversal of grated surfaces and grated stairs
- o Integrated a deterministic finite state machine for blind locomotion, eliminating low-level crashes across months of subsequent field testing
- o Designed filtering and tracking pipelines for quadruped mounted 6 DOF manipulator, removing drift and reducing end-effector tracking error

Optimal Robotics Lab

2018 - 2023

Graduate Research Assistant (Matlab & Python)

- o Obstacle avoidance algorithm: for quadrotor & bipedal avoidance of fast-moving obstacles
[hardware, ~1kHz] <https://www.youtube.com/watch?v=cKKLuRdrqcQ>
- o Bipedal balancing controller: for dynamic terrain tested on Cassie
[hardware, >2kHz] <https://www.youtube.com/watch?v=q10Wg61H4UQ>
- o Real-time reactive planning and control for monopods (with flexible contact sequencing)
[simulated, ~500Hz] <https://www.youtube.com/watch?v=EquyU9cm-SM>
- o Quadrupedal stepping controller: using optimal control techniques (preliminary results)
[simulated, ~500Hz] www.linkedin.com/feed/update/urn:li:activity:7122585426162249730/

Technical Skills

Languages: C++, Python, MATLAB

Simulations: MuJoCo, Simulink, Bullet

Control Methods: Operational Space Control, Whole-Body Control, Forward/Inverse Dynamics, Joint Space Control, PD/PID Control, Force Control

Planning: Model Predictive Control, Direct Collocation, Sampling-based methods (RRT*, A*)

Optimization: Linear/Quadratic/Nonlinear Programming, Newton-Raphson

Awards, Fellowships, and Honors

2022: NASA Big Idea Challenge Finalist (ET-QUAD)

NASA

2022: Best Locomotion Paper, IEEE International Conference on Robotics and Automation

IEEE

2021: Research Fellowship Recipient, DEVCOM Army Research Laboratory

National

Peer Reviewed Publications

Tianze Wang, **Jason White**, and Christian Hubicki, "Real-time Dynamic Bipedal Avoidance," in 2023 *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* IEEE.

Jason White, David Jay, Tianze Wang, and Christian Hubicki, "Avoiding Dynamic Obstacles with Real-time Motion Planning using Quadratic Programming for Varied Locomotion Modes," in 2022 *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (pp. 13626-13633) IEEE.

Adwait Mane, Dylan Swart, **Jason White**, and Christian Hubicki. "Trajectory Optimization Formulation with Smooth Analytical Derivatives for Track-leg and Wheel-leg Ground Robots." in 2022 *International Conference on Robotics and Automation (ICRA)* (pp. 5762-5768). IEEE.

Jason White, Max Austin, Jonathan Clark, Christian Hubicki, and Jason Pusey. "Online Gait Optimization for Running in Resistive Media," in 2022 *7th International Conference on Robotics and Automation Engineering (ICRAE)* (pp. 282-286). IEEE.

Max Austin, John Nicholson, **Jason White**, Sean Gart, Ashley Chase, Jason Pusey, Christian Hubicki, Jonathan Clark, "Optimizing Dynamic Legged Locomotion in Mixed, Resistive Media," in 2022 *IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* (pp. 1482-1488). IEEE.

Jason White, Dylan Swart, and Christian Hubicki, "Force-based Control of Bipedal Balancing on Dynamic Terrain with the "Tallahassee Cassie" Robotic Platform," in 2020 *IEEE International Conference on Robotics and Automation (ICRA)* (pp. 6618-6624). IEEE.

Manuscripts & Posters

Jason White and Christian Hubicki, "Through-contact Monopod Motion Generation via MPC: Convex Formulation and Locomotion Analysis." 2024.

Jason White and Christian Hubicki, "Real-time Adversarial Obstacle Avoidance." 2023.

Jason White and Christian Hubicki "Toward Reduced-Order Modeling Approaches for Rapidly Reactive Locomotion." 2021 Dynamic Walking

Education

Florida State University

Advisor: Christian Hubicki Ph.D.

Ph.D. Mechanical Engineering - Legged Robot Control

M.S. Mechanical Engineering - Robotics, Dynamics, & Controls

Virginia Polytechnic Institute and State University

B.S. Civil Engineering - Structural Focus